Función de transferencia

$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\, dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de laplace

$$F_a(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(t) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_c(t) = Cs\,P_p(s)$$

$$F_R(t) = \frac{P_p(s)}{R}$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

Procedimiento algebraico

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs\,P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{Ls}\right) P_a(s) = \left(Cs + \frac{1}{R} + \frac{1}{Z} + \frac{1}{Ls}\right) P_p(s)$$

$$\frac{Ls+Z}{LZs}\,P_a(s) = \frac{CLRZs^2 + LZs + LRs + RZ}{LRZs}\,P_p(s)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{LRZs}}$$

• Resultado •

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}$$

Error en estado estacionario

Este error se calcula la entrada [$P_a(s)$] es un escalón unitario, es decir $P_a(s) = 1/s$;

$$e(s) = \lim_{s\to\infty} s\,P_a(s)\left(1 - \frac{P_p(s)}{P_a(s)}\right) = \lim_{s\to\infty} s\frac{1}{s}\left(1 - \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}\right) = \emptyset$$

$$e(t) = \emptyset V$$

Estabilidad del sistema en lazo abierto

Para analizar la estabilidad del sistema en lazo abierto se deben calcular los polos;

$$CLRZs^2 + L(Z+R)s + RZ = 0$$

$$a = CLRZ$$

$$b = L(Z+R)$$

$$c = RZ$$

$$\lambda_{1,2} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2CLRZ}$$

Hipotenso

$Z = 0.010$	$P_p(s) = \frac{LRs + RZ}{Pa(s)} = \frac{0.001s + 0.01}{CLZs^2 + L(Z+R)s + RZ} = \frac{0.001s + 0.01}{3.5\times10^{-5}s^2 + 0.0101s + 0.01}$
$L = 0.001$	
$R = 1.000$	$\lambda_1 = -14.429 + 8.805\,1i$
$C = 3.500$	$\lambda_2 = -14.429 - 8.805\,1i$

por lo tanto el sistema presenta una respuesta estable y subamortiguada

Normotenso

$Z = 0.010$	$\frac{P_p(s)}{Pa(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ} = \frac{0.001s + 0.01}{1.5\times10^{-5}s^2 + 0.0101s + 0.01}$
$L = 0.001$	
$R = 1.000$	$\lambda_1 = -12.062,$
$C = 1.500$	$\lambda_2 = -55.272,$

$Z = 0.010$	$R = 1.000$
$L = 0.001$	$C = 1.000$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ} = \frac{0.001s + 0.01}{0.00001s^2 + 0.0101s + 0.01}$$

$$\lambda_1 = -10.0,$$

$$\lambda_2 = -10.0,$$

Modelo de ecuaciones integro-diferenciales

$$F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t) - P_p(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt = C\left(\frac{dP_p(t)}{dt}\right) + \frac{P_p(t)}{R}$$

Nota: la única variable desconocida es la salida del sistema, $P_p(t)$
NO se puede despejar de términos en los que se esté derivando o integrando.

$$\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{Z}$$

$$P_p(t)\left(\frac{1}{R} + \frac{1}{Z}\right) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p}{dt}\right]$$

$$P_p(t)\frac{Z+R}{RZ} = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p}{dt}\right]$$

$$P_p(t) = \left[\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\left(\frac{dP_p(t)}{dt}\right)\right)$$

$$\frac{RZ}{Z+R}$$